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Artificial intelligence (A.I) and Precision farming have been proposed as solutions to improve agricultural productivity in India. Critically analyze their feasibility and challenges in Indian agricultural landscape.

India's finance minister Nirmala Sitharaman has termed agriculture as the first and most crucial growth engine of Indian Economy.

It is not only a traditional practice sector but it is also important for India's.

- 1) Food Security
- 2) Rural economy
- 3) Self reliance

Therefore budget 2025-26 has seen

major policy shift and a new investment road map which will help agriculture to

- 1) Modern Agriculture
- 2) Increase productivity.

this will help in increasing productivity of crop yielding by using Artificial intelligence

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and precision farming.

Precision farming :->

It is also known as precision agriculture, a technique that uses technology to increase crop yield and improve management by using sensors, Drones, satellite mapping, data analytics tools.

Feasibility Areas

- 1) land
- 2) labour
- 3) Electrification in rural areas.
or
Infrastructure.
- 4) Economic aspect / credit
- 5) Digital literacy.

challenges

Consolidation of land is still in progress.

skilling of labourers

electrification is still under progress in many rural areas.

as most of our farmers base is lacking in initial capital investment.

lacking in necessary

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digital skill to operate and interpret A.I based system.

Other than this the challenges in following precision agriculture is about

- 1) Institutes → less number of Agriculture institute to ~~are~~ analyse data and study agriculture Map starts.
- 2) Ex:- ICAR, Agriculture universities, Krishi Bhavan, IARI (Indian Agri Research Institute)

*) government policies and scheme to overcome Challenges are :-

- 1) PM Dhan - Dhanya Krishi Yojana :-

- It is inspired from aspirational district program to increase productivity in low low productivity agricultural district.
- using A.I based precision farming,
- micro irrigation

NITI Ayog report $\Rightarrow$ 85% of Indian farmers have land holding less than 2 hectare. So for A.I and precision Agriculture land consolidation is a problem.

2) High yielding seeds $\Rightarrow$ [National Mission]

Introducing high grade quality seeds to improve yielding in production.

So, precision agriculture institutes will help in research development in,

- 1) Climate resistant crop seeds.
- 2) Disease free seeds.
- 3) Nutrients level will be more to tackle nutritional aspects of India in fighting against hunger (SDG-1, 2).

3) Irrigation aspect $\Rightarrow$

As 60% of Indian irrigated land depend on traditional irrigation, pump set, tubewell, canals, so water wastage leads to low crop productivity.

Micro irrigation system or drip irrigation system will ensure more productivity.

govt scheme: → "HAR KHET TAK PAANI" scheme.

↳ to fight drought

2) Rashtriya Krishi Vikas Yojana

- to focus more on effectively use of water through drip and sprinkler irrigation.

use of precision farming will help in improving agriculture productivity. with the upskilling of the farmers through Digital literacy programme, and providing initial economical support to shift from traditional to Artificial intelligence in agriculture practice will help in improving agricultural productivity and will ultimately help in achieving India Sustainable development goal of No poverty, zero hunger. and it will also place India among the top crop yielding with ~~top~~ quality. countries.